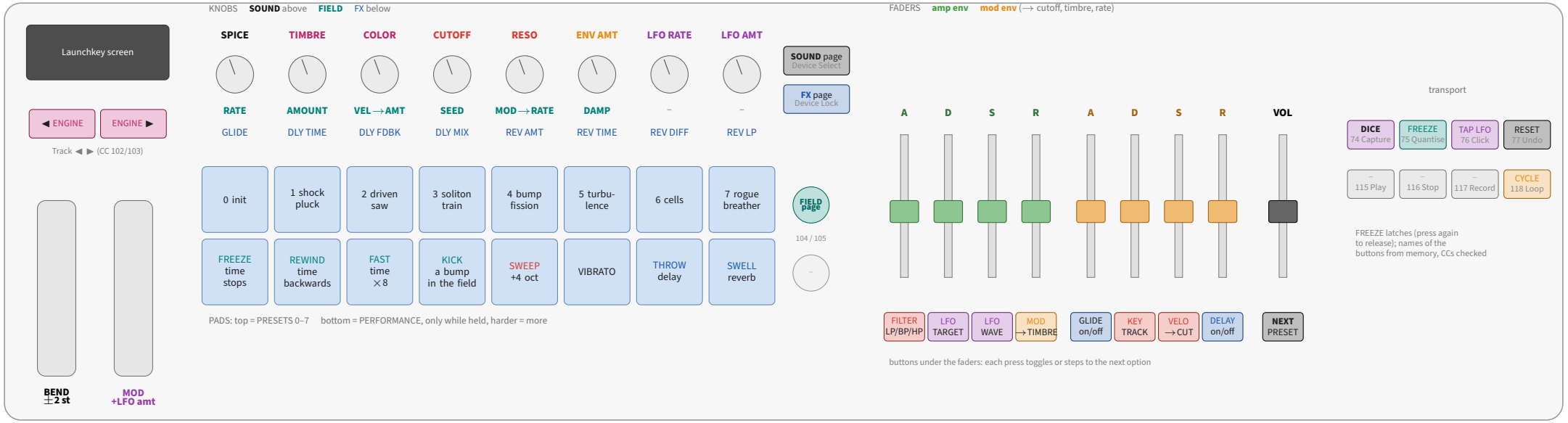
soliton

Axoloti Core + Launchkey 49 MK3 (standalone mode) · 3 voices, each a field under a nonlinear PDE · `patches/soliton/soliton_v1.axp`

github.com/ndlopes-github/axoloti-playground



Engines (Track ◄ / ►): TIMBRE and COLOR per engine

SHOCK a sine steepens into a shock (a saw), brighter, then decays as the shock eats its energy: the physics is the envelope. TIMBRE: sharper (less viscosity). COLOR: DRIVE, a push that keeps a steady saw going.

SOLITON a wave breaks into solitons that overtake each other and nearly come back together. TIMBRE: more solitons (less dispersion). COLOR: the start, a cosine → a single bump (it splits into a train).

CHAOS cells that drift, split and merge, never quite repeating. TIMBRE: room for more cells: steady → travelling → chaotic, and brighter. COLOR: dispersion, which tames the chaos into regular pulses.

ROGUE a wave group on a carrier: sidebands grow, a peak builds, it falls back, again. TIMBRE: more unstable sidebands. COLOR: the carrier harmonic, 1–5.

Changing engine while a note sounds: SHOCK, SOLITON and CHAOS take over the field as it is (one equation's state goes on under another); to or from ROGUE the note starts again.

FIELD page (round button, upper)

RATE how fast the field's time runs, 0.05 – 20 units/s (exponential). Above the engine's speed limit (page 2) it plays at the limit.

AMOUNT the size of a new note's field: how nonlinear it is. Louder fields steepen, split and focus sooner.

VEL→AMT velocity scales AMOUNT, $\times\frac{1}{2}$ – ×2 at full.

SEED the perturbation a note starts with (CHAOS and ROGUE grow from it).

MOD→RATE centre = none; right: the mod envelope speeds time up (×16 at full), left: slows it.

DAMP damping, 0 – 2 per unit: the field relaxes; the level is held, so it is heard as a simpler, purer tone.

Keys: 3 voices, one field each

Each note starts its own field, at the key's pitch: the field is the waveform, one period of it one cycle, so tuning is exact at any timbre. Velocity sets the field's size (VEL→AMT), so it changes the *evolution*, not only the level. A chord is three fields evolving side by side.

Time

The timbre moves because the field's time runs. FREEZE (pad, pressure slows it to a stop, or the Quantise latch) holds the timbre; FAST speeds it up to ×8; REWIND runs it backwards, faster with pressure. Only SOLITON and ROGUE can run backwards: SHOCK and CHAOS dissipate, and running them backwards is ill-posed, so for them REWIND freezes. KICK drops a bump into each sounding field (SOLITON: a new soliton; ROGUE: a focusing seed).

Envelopes, filter, LFO

As freak: faders 1–4 amp, 5–8 mod (1 ms – 10 s; S 0 – 1); CYCLE (Loop) loops the mod env. Mod env → cutoff (ENV AMT), → TIMBRE (+0.5, MOD→TIMBRE) and → RATE (MOD→RATE). LFO TARGET: timbre (±0.5) → rate ($\times\frac{1}{4}$–4) → cutoff (±3 oct) → pitch (±2 st). Filter: 12 dB/oct SVF, LP/BP/HP.

Pads

Top = presets 0–7. **Bottom** = performance, while held; strike velocity, then pressure, sets the amount:

FREEZE	time slows to a stop
REWIND	time backwards ($\times\frac{1}{2}$ – ×2)
FAST	time up to ×8
KICK	one bump per strike
SWEEP	cutoff +4 oct
VIBRATO	6 Hz, ±0.7 st
THROW / SWELL	delay throw / reverb swell

SPICE

Per note: the seed's phases, pitch (±0.5 st), TIMBRE (±0.2), time rate (±40 %) and cutoff (±1.5 oct). In CHAOS two notes never evolve alike.

The equations ($y \in [0, 2\pi)$, periodic)

SHOCK, Burgers: $u_t + uu_y = \nu u_{yy} + F \sin y$

SOLITON, KdV: $u_t + uu_y + \delta^2 u_{yyy} = 0$

CHAOS, Kuramoto–Sivashinsky on a domain of length L ($s = 2\pi/L$): $u_t + suu_y + s^2 u_{yy} + \beta s^3 u_{yyy} + s^4 u_{yyyy} = 0$

ROGUE, focusing NLS: $i\psi_t + s^2\psi_{yy} + 2|\psi|^2\psi = 0$, heard as $\mathrm{Re}(\psi e^{imy})$

Fourier pseudo-spectral, 128 points, 42 modes (2/3 rule); the linear part exact, Strang splitting with RK4 (Cayley rotation for NLS); the waveform is the field, normalised to a fixed RMS (SHOCK: to its start, so its decay is heard), band-limited for its pitch. Details: `lib/soliton_field.h`.

Signal path and load

3 × (field → SVF → amp env × velocity) → delay → Elements reverb → VOL. Each field takes 125 steps/s. DSP 38 % idle, 61.5–68.5 % with 3 notes (`patches/bench_soliton`); patch RAM 36 of 44 KB.

Presets: top pads = 0–7; NEXT PRESET (fader button 9) steps; RESET (Undo) = 0

(chosen from the parameter meanings, not yet tuned by ear)

0 init	SOLITON, a few solitons from a cosine; time 2.2 units/s; LP 8 kHz. The power-on sound.
1 shock pluck	SHOCK, sharp, no drive, loud fields: the sine steepens into a saw in about 0.3 s while the amp envelope decays (0.7 s); delay on. Play it short.
2 driven saw	SHOCK with COLOR (DRIVE) at 0.6, $F = 0.7$: a steady saw that keeps reshaping; slow attack, pad envelope.
3 soliton train	SOLITON at the most solitons, RATE 5.6 units/s (the speed limit holds louder notes to 1–4): the brightest motion of the set.
4 bump fission	SOLITON from a single bump that splits into a train; MOD→RATE: fast at the attack, slow after.
5 turbulence	CHAOS, wide domain (about 10 cells): starts from order and turns chaotic within a second or two; LFO on the rate (sample & hold); delay on.
6 cells	CHAOS, narrow domain (3–4 cells) with dispersion: steady, slowly drifting pulses.
7 rogue breather	ROGUE, 1 unstable sideband on carrier 1: the tone swells into a peaked, brighter wave and back, again and again.

A preset sets the engine, all three knob pages, faders 1–8 and the toggles; the volume stays. **DICE** (Capture MIDI) rolls the engine and the SOUND and FIELD pages instead; envelopes, FX and volume stay.

Speed limits: how fast the field's time can run (units/s, RATE at full)

The step that keeps each equation stable and accurate shrinks as the field grows, so loud notes evolve more slowly at the top of RATE: SHOCK 16–18 (less while the shock forms) SOLITON 3.9 (AMOUNT 1/2) – 0.9 (the loudest notes) CHAOS about 10 ROGUE 3. Below these, RATE is exact.

Sound design moves

Frozen timbre: let a note evolve, then latch FREEZE (Quantise): a static wave you chose by ear. **Scrub the evolution**: alternate FAST and REWIND on SOLITON. **Plucked shock**: SHOCK, TIMBRE full, short amp decay, velocity on (VEL→AMT full): harder = brighter sooner. **Living drone**: CHAOS, TIMBRE 3/4, slow RATE, long release; SPICE 1/3. **Strike a soliton**: SOLITON, hold a note, KICK it. **Order to chaos**: CHAOS, MOD→RATE left: fast chaos at the attack that settles.

Where the values fall (knob turned . . .)

	1/4	1/2	3/4	full
RATE (time units/s)	0.23	1.0	4.4	20
DAMP (per unit)	0.13	0.51	1.1	2
SHOCK TIMBRE: viscosity ν	0.105	0.037	0.013	0.0047
SHOCK COLOR: drive F	0.13	0.51	1.1	2
SOLITON TIMBRE: dispersion δ	0.33	0.22	0.15	0.10
CHAOS TIMBRE: $L/2\pi$ (cells)	3.9	6.6	9.3	12
CHAOS COLOR: dispersion β	0.25	1.0	2.2	4
ROGUE TIMBRE: unstable sidebands (field size 1)	1	2	3	4
ROGUE COLOR: carrier harmonic	2	3	4	5

TIMBRE at 0: ν 0.3, δ 0.5, $L/2\pi$ 1.1, 1 sideband; ROGUE carrier 1 at COLOR 0. SOLITON's δ stops at $0.1\sqrt{A}$ for loud fields (A: field size), so the solitons stay resolved. Envelopes, CUTOFF, RESO, ENV AMT, LFO, GLIDE, delay and VOL fall as on the freak card.

Something seems wrong?

No sound: VOL up? CUTOFF as on freak.

Nothing moves: FREEZE latched (Quantise); RATE at the bottom; REWIND held on SHOCK or CHAOS (it freezes them); SHOCK has decayed (raise DRIVE).

Moves too slowly: a loud SOLITON note is at its speed limit: lower AMOUNT or VEL→AMT.

Dull low notes: a field has 42 harmonics, so C2 reaches 2.7 kHz at most: open the filter, or play higher.

Lost: RESET (Undo).

MIDI reference (beyond the freak card)

FIELD page (round button, upper)	ch 16 CC 104
ENGINE − / + (track ◀/▶)	ch 16 CC 102 / 103
FREEZE latch (Quantise)	ch 16 CC 75
FREEZE, REWIND, FAST, KICK (pads)	ch 10 notes 36–39
free: round button lower, Play, Stop, Record	ch 16 CC 105, 115–117